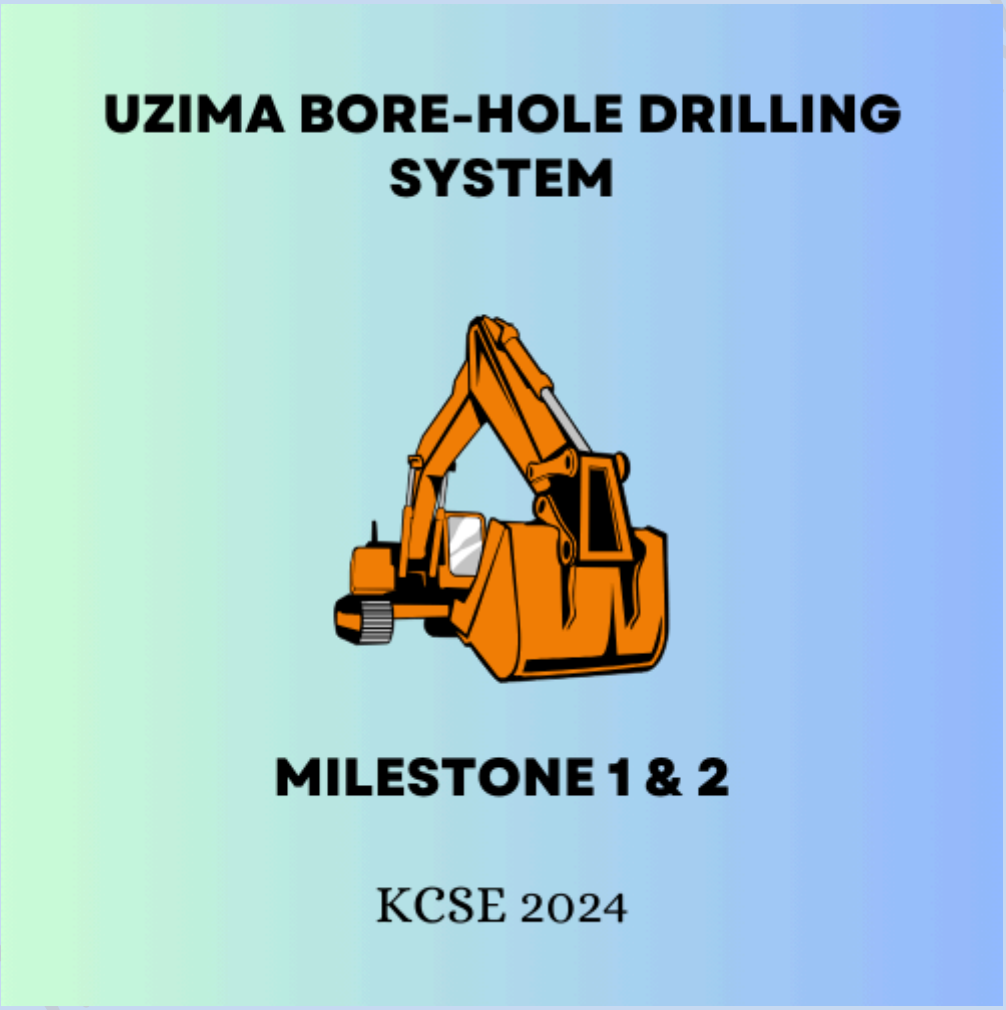
UZIMA BOREHOLE SYSTEM

KENYA CERTIFICATE OF SECONDARY EDUCATION

DOCUMENTATION

KCSE COMPUTER STUDIES PROJECT 2024 (451/3)



UZIMA BOREHOLE SYSTEM

NAME: /……

SCHOOL CODE: ` …… /……

INDEX NUMBER ……

PROJECT TITLE: UZIMA BORE HOLE SYSTEM

YEAR OF EXAMINATION: 2024

SCHOOL: /……..

UZIMA BOREHOLE SYSTEM

DECLARATION

I vow that this project is my original work. I have not copied anywhere directly from the internet or any other source. It is a result of my own effort through consultation from my computer teacher. This work has also never been presented to the Kenya National Examination Council.

NAME SIGNATURE

MR……I (SUPERVISOR) …….....

……..

UZIMA BOREHOLE SYSTEM

@2023PROJECT

ACKNOWLEDGEMENT

I would like to acknowledge all those who have made this project a success. For a project like this, I did not think I would have come up with it without their support. I would like to thank everybody who was with me throughout this journey.

I would also like to acknowledge the efforts of my teacher, who was responsible for my training and enabled me to get to know much about computers and window application programs which has enabled me do this project.

Lastly, I would like to thank my school mates, most especially my classmates who have provided me with suggestions concerning this project.

DEDICATION

I also take this chance to dedicate this project to anyone interested in Microsoft Access.

UZIMA BOREHOLE SYSTEM

Tables of contents

KENYA CERTIFICATE OF SECONDARY EDUCATION..... i
DOCUMENTATION..... i
KCSE COMPUTER STUDIES PROJECT 2024 (451/3)..... i
NAME: /……..... ii
SCHOOL CODE: ` …….. /……..... ii
INDEX NUMBER ……..... ii
PROJECT TITLE: UZIMA BORE HOLE SYSTEM..... ii
YEAR OF EXAMINATION: 2024..... ii
SCHOOL: /……..... ii
DECLARATION..... i
ACKNOWLEDGEMENT..... ii
DEDICATION..... ii
Tables of contents..... 3
Background Information..... 1
Introduction..... 1
Overview of the existing system..... 3
Inefficiencies in the Current System..... 3
Data Processing Challenges..... 3
Issues with the Manual Approach..... 3
Proposed Computerized System..... 3
Objectives..... 3
Demerits of the current system..... 3
Organizational structure:..... 5
Problem Statement:..... 6
Overview of the proposed system..... 8
Objectives..... 10
Advantages..... 12
Feasibility study..... 14
Feasibility Report..... 14
Cost benefit analysis..... 14
BENEFITS..... 15
Schedule feasibility..... 15

UZIMA BOREHOLE SYSTEM

Economic feasibility..... 15
Operational feasibility..... 16
Technical feasibility..... 16
Conclusion..... 16
FACT FINDING REPORTS..... 18
Documentation study..... 18
Questionnaire..... 18
UZIMA BORE HOLE SYSTEM SPORT CLUB MANAGEMENT SYSTEM..... 19
Interviews..... 20
Observations..... 21
Recommendation..... 22
SYSTEM DESIGN: System flowcharts..... 24
1.0 CLIENT REGISTRATION..... 24
1.1 Borehole Drilling Project Management..... 25
1.2 Financial Transaction Processing..... 26
1.3 System Replacement Decision..... 26
1.4 SALE ITEM..... 27
1.5 RESTOCK ITEMS..... 28
1.6 EQUIPMENT LOSS AND DAMAGE..... 29
OUTPUT FLOWCHAT..... 30
2.3 31
2.4 31
2.5..... 32
2.6..... 33
2.7..... 33
FIGURE 7..... 33
FORMS..... 34
Figure 3.0..... 34
Figure 3.1..... 34
Figure 3.3 Item Registration Form..... 35
Figure 3.4 35
4.3 OUTPUT DESIGN..... 36
Figure 19 Items Table Report..... 36
Figure 20..... 36

UZIMA BOREHOLE SYSTEM

Figure 21..... 37
Figure 22..... 37
Figure 23..... 39
Figure 24..... 39
4.5 HARDWARE AND SOFTWARE REQUIREMENTS..... 40
4.7 DATA INTERGRITY CONSTRAINTS..... 42
4.8.1 Entity integrity:.....42
Referential integrity:..... 43
Figure 23 Relationships.....44
SYSTEM IMPLEMENTATION.....45
Staff training.....45
Demonstrations:.....45
Workshops:..... 45
Manuals:..... 45
Changeover strategy..... 46
Advantages:.....46
Disadvantages:..... 46
File conversion..... 47
File Conversion in System Implementation:.....47
Operating System Enhancement:.....47
Software Replacement Approach:..... 47
6.1 USER MANUAL..... 49
6.1.1 System installation..... 49
HOW TO INSTALL THE SYSTEM ON YOUR COMPUTER:..... 49
6.1.2 Running the interface.....50
FIGURE 26..... 50
6.1.4 Error handling..... 51
7.0 BIBLIOGRAPHY..... 52

UZIMA BOREHOLE SYSTEM

UZIMA BOREHOLE SYSTEM

CHAPTER ONE

Background Information

Introduction

The demand for clean and reliable water supply is increasing in Kenya, especially in urban and rural areas where access to water sources is limited or unreliable. Borehole drilling is one of the methods that can provide water to households, businesses, and industries. However, borehole drilling is a complex and costly process that involves various factors such as the type of drilling, the depth and height of the borehole, the type of pump and tank, and the fees and taxes involved.

Uzima Company is one of the leading borehole service providers in Kenya, offering borehole drilling, pump installation, tank installation, plumbing, and pump maintenance services to industrial, commercial, and domestic clients. The company charges different rates for different services, depending on the client category, the type of drilling, the type of pump, the depth and height of the borehole, the capacity of the tank, the type and length of the pipes, and the number of outlets. The company also collects survey fees and local authority fees from the clients, as well as 16% tax on the total cost of the service.

The aim of this project is to develop a well-documented computerized database system that would help Uzima Company to manage its operations more efficiently and effectively. The system would have the following features:

- Register and maintain a list of clients, with their details such as name, address, telephone number, and intended borehole location.
- Register items and services provided by the company, with their details such as type, cost, and quantity.
- Compute the total amount to be paid by each client, based on the services and items they requested, the fees and taxes they incurred, and the discounts they received.
- Compute the total amount charged on plumbing, based on the type, diameter, length, and number of pipes used.
- Compute the tax paid by each client, based on the 16% tax rate applied on the total cost of the service.
- Compute the total fees charged for survey and local authority, based on the client category and the fees
- Compute the total revenue generated by the company from each service from all clients, based on the total amount paid by each client for each service.
- Generate appropriate reports, such as invoices, receipts, statements, and summaries, that would show the details and calculations of the above features.

UZIMA BOREHOLE SYSTEM

@2024PROJECT

The system would be developed using Microsoft Access as the database management system, and Visual Basic as the programming language. The system would use a relational database model, with tables, queries, forms, reports, and macros. The system would also have a user-friendly interface, with menus, buttons, and validations.

UZIMA BOREHOLE SYSTEM

@2024PROJECT

Overview of the existing system

To become a member of the Uzima Bore Hole System sport club, individuals are required to provide personal details such as names, ID number, contact information, and additional information.

Inefficiencies in the Current System

The existing system faces significant inefficiencies:

- All records are manually written on papers and counter books.
- The use of counter books occupies a considerable amount of space in the offices.
- Papers are susceptible to damage over time.
- Retrieving youth records is time-consuming.
- Registration records are maintained in a counter book, and each member has an individual counter book for monthly contributions.

Data Processing Challenges

The council relies heavily on manual data processing for various activities, including:

- Member registration
- Monthly contribution recording
- Managing inventory
- Facilitation fees tracking
- Stock levels monitoring
- Sales recording

Issues with the Manual Approach

The manual data processing approach results in:

- Increased likelihood of errors in data entry and retrieval.
- Difficulty in managing and accessing records efficiently.
- Limited scalability and potential loss of data due to paper damage.
- Inefficiencies in handling inventory, facilitation fees, stock levels, and sales.

Proposed Computerized System

Objectives

The primary objectives of implementing a computerized system are:

- Streamlining the registration process for members.
- Improving data accuracy and reducing errors.
- Efficiently managing and organizing records.
- Enhancing inventory and sales management.

UZIMA BOREHOLE SYSTEM

Demerits of the current system

After a thorough analysis of the manual system used for managing Uzima Borehole Drilling operations, several shortcomings have been identified:

Ineffective Record Maintenance:

- ✔ The current manual system struggles to efficiently maintain accurate records of clients, borehole drilling projects, and associated services, leading to potential errors and oversights.

Challenges with Handling Increased Workload:

- ✔ As the number of borehole drilling projects and clients grows, the limitations of the current system become apparent, impacting its ability to effectively process and manage orders.

Inaccuracy in Processing Borehole Drilling Records:

- ✔ The manual data entry process is prone to errors, resulting in inaccuracies in the records of borehole drilling services, pump installations, and plumbing activities.

Data Entry Errors:

- ✔ Manual data entry increases the risk of errors, including typos, incorrect calculations, and inconsistencies in recording information, potentially leading to misunderstandings and client dissatisfaction.

Delayed Report Production:

- ✔ The manual generation of reports is a time-consuming process, causing delays in obtaining critical insights into the company's performance, client payments, and project statuses

High Overall Operating Costs:

- ✔ The reliance on manual processes contributes to higher operational costs, including increased labour expenses, as more resources are required to manage the growing volume of borehole drilling projects and associated services.

Data Integrity and Computational Errors:

- ✔ The manual nature of the system introduces the risk of data integrity issues and computational errors, particularly in the calculation of fees, taxes, and revenue generated from borehole drilling services.

Security Concerns:

- ✔ The current system poses security risks as sensitive data, including client details, financial information, and project specifics, is vulnerable to unauthorized access, theft, or tampering.

In light of these demerits, it becomes imperative to transition to a computerized database system to enhance efficiency, accuracy, and security in managing Uzima Borehole Drilling operations. The proposed system aims to address these challenges and provide a more streamlined and secure approach to data management and analysis.

UZIMA BOREHOLE SYSTEM

Organizational structure:

Managing Director:

- ✔ Responsible for overseeing the overall operations and strategic direction of Uzima Borehole Drilling Company. Ensures the company's mission and goals are achieved.

Assistant Director (Operations):

- ✔ Assists the Managing Director in day-to-day operations, including project management, client relations, and team coordination. Takes charge in the absence of the Managing Director.

Finance Manager:

- ✔ Manages financial aspects, including budgeting, accounting, and financial reporting. Ensures proper allocation of funds for drilling projects and other services.

Project Coordinators:

- ✔ Oversee specific drilling projects from initiation to completion. Coordinate with drilling teams, clients, and other stakeholders to ensure project success.

Technical Supervisor:

- ✔ Manages and supervises the technical aspects of drilling operations. Ensures adherence to safety standards, drilling procedures, and equipment maintenance.

Secretary and Records Keeper:

- ✔ Responsible for documenting meeting minutes, maintaining records of drilling projects, and managing administrative tasks to ensure smooth communication and organization within the company.

Client Relations Manager:

- ✔ Handles client interactions, addresses inquiries, and ensures client satisfaction. Facilitates the application and registration process for clients seeking borehole drilling services.

Health and Safety Officer:

- ✔ Focuses on promoting and enforcing safety measures in all drilling activities. Monitors compliance with health and safety regulations.

Borehole Drilling Teams (Captains and Team Members):

- ✔ Headed by experienced Captains, these teams lead drilling operations in the field. Team members work together to carry out drilling tasks efficiently.

Support Staff (Administrative, Technical, and Field):

- ✔ Includes administrative assistants, technical support staff, and field assistants who contribute to the smooth functioning of the company.

UZIMA BOREHOLE SYSTEM

Problem Statement:

The current manual system utilized by Uzima Company for managing its borehole drilling operations presents several inefficiencies and challenges that hinder operational efficiency, data accuracy, and scalability. These challenges include:

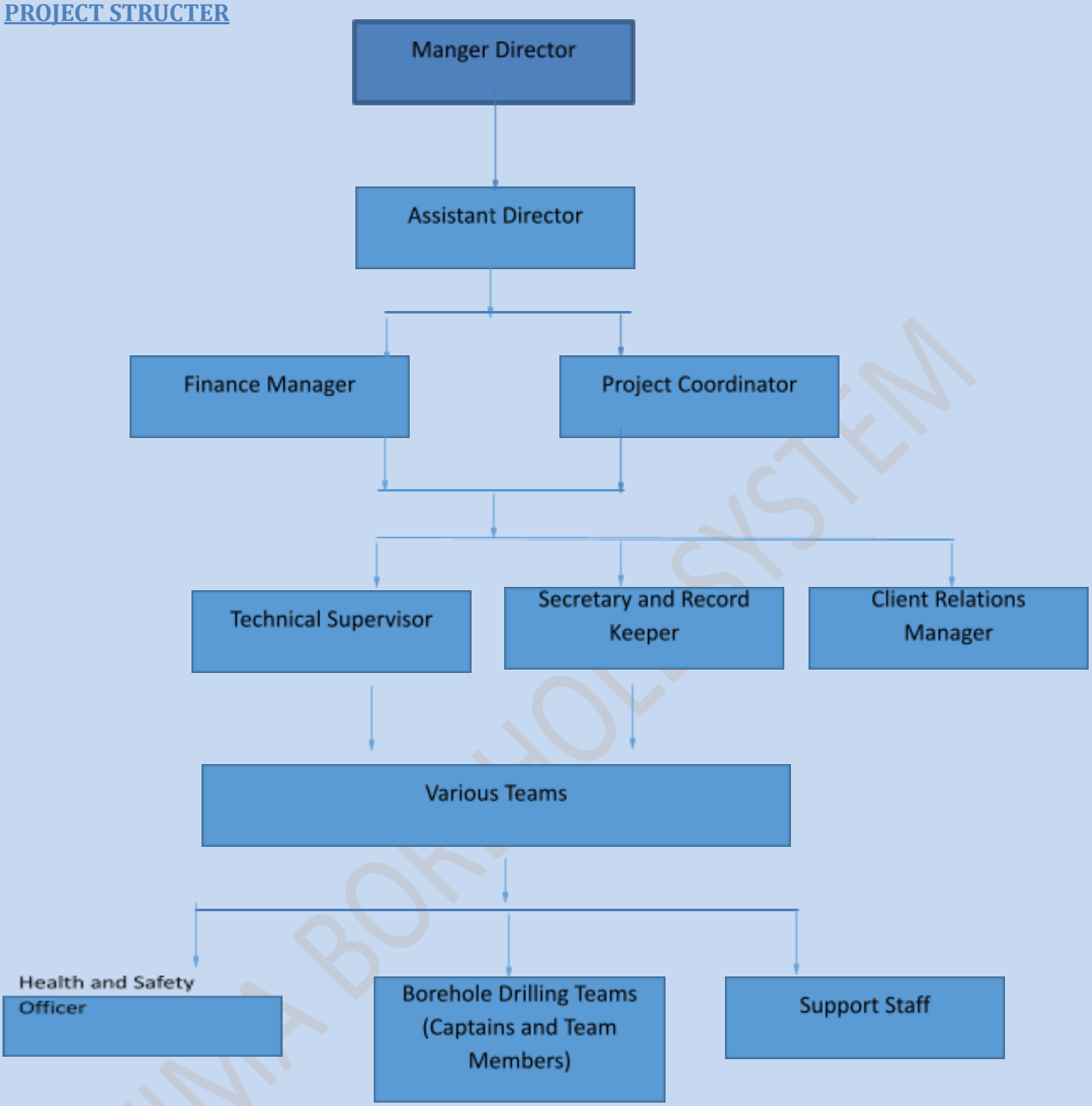
1. **Ineffective Record Maintenance:**
   The manual system struggles to maintain accurate records of clients, borehole drilling projects, and associated services, leading to potential errors and oversights in data management.
2. **Challenges with Handling Increased Workload:**
   As the number of borehole drilling projects and clients grows, the limitations of the current system become apparent, impacting its ability to effectively process and manage orders, resulting in delays and potential customer dissatisfaction.
3. **Inaccuracy in Processing Borehole Drilling Records:**
   Manual data entry processes are prone to errors, leading to inaccuracies in the records of borehole drilling services, pump installations, and plumbing activities, which can affect billing accuracy and financial reporting.
4. **Data Entry Errors:**
   Manual data entry increases the risk of errors such as typos, incorrect calculations, and inconsistencies in recording information, leading to misunderstandings and client dissatisfaction.
5. **Delayed Report Production:**
   The manual generation of reports is time-consuming, causing delays in obtaining critical insights into the company's performance, client payments, and project statuses, which hampers decision-making and planning processes.
6. **High Overall Operating Costs:**
   Reliance on manual processes contributes to higher operational costs, including increased labor expenses and potential data loss due to paper damage, limiting the company's profitability and growth potential.

7. **Data Integrity and Security Concerns:**
   The manual nature of the system introduces the risk of data integrity issues, computational errors, and security breaches, as sensitive information is vulnerable to unauthorized access, theft, or tampering, posing risks to client confidentiality and company reputation.

Addressing these challenges requires the implementation of a computerized database system that streamlines operations, enhances data accuracy, improves reporting capabilities, and strengthens data security measures, enabling Uzima Company to optimize its borehole drilling services, increase customer satisfaction, and maintain competitiveness in the market.

UZIMA BOREHOLE SYSTEM

PROJECT STRUCTER



UZIMA BOREHOLE SYSTEM

@2024PROJECT

Overview of the proposed system

The proposed system for Uzima Borehole Drilling is a comprehensive database designed to streamline and enhance the management of various aspects within the organization. The system is tailored to meet the specific needs of a borehole drilling company, ensuring efficiency and accuracy in record-keeping and financial computations. Here's an overview of the key functionalities:

Member Registration:

- ✔ The system will facilitate the registration of new clients seeking borehole drilling services. This includes capturing essential client details such as name, address, contact information, and intended borehole location.

Clients Records:

- ✔ The database will maintain records of individuals involved in borehole drilling teams and various organizational activities. This includes details such as roles, responsibilities, and participation in drilling projects.

Services Records:

- ✔ Comprehensive records of borehole drilling projects, pump installations, plumbing services, and other related activities will be stored in the system. This ensures a detailed overview of the services provided by Uzima Borehole Drilling.

Inventory Management (Items and Stock Levels):

- ✔ The system will keep track of inventory, including items such as drilling equipment, pumps, and plumbing materials. It will monitor stock levels, alerting when re-stocking is necessary to ensure smooth project execution.

Patron Information:

- ✔ Details about patrons who facilitate teams, games, or other services will be recorded. This includes their contact information and involvement in supporting the organization's activities.

UZIMA BOREHOLE SYSTEM

Financial Computations:

- ✔ The system will compute various financial aspects, including payments for services, item sales, re-stocking expenses, total revenue generated by the company, and the overall profit. It will consider factors such as drilling fees, pump installation charges, plumbing service costs, and taxes.

Commission Tracking:

- ✔ If applicable, the system can track commissions for individuals or teams involved in sales or services. This helps in transparently managing compensation for different roles within the organization.

Reports Generation:

- ✔ The proposed system includes a reporting feature to generate comprehensive reports. These reports may include financial summaries, project progress, inventory status, and other key performance indicators.

The overall goal of the proposed system is to replace the current manual processes with an efficient and reliable computerized database. This will not only improve record-keeping accuracy but also enhance the company's ability to make informed decisions based on real-time data. The system aims to optimize organizational processes, reduce errors, and contribute to the overall success and growth of Uzima Borehole Drilling Company.

UZIMA BOREHOLE SYSTEM

Objectives

User-Friendly Data Entry Forms:

- ✔ Design and implement intuitive forms within the system to facilitate the easy entry of information, such as youth registration details. These forms should provide a user-friendly interface for capturing necessary data accurately and efficiently.

Efficient Record Processing:

- ✔ Develop robust processes to handle and maintain records of youths, drilling teams, and inventory management. Ensure that the system can effectively process and update information related to borehole drilling projects, pump installations, plumbing services, and other relevant activities.

Comprehensive Youth Registration:

- ✔ Create dedicated forms and functionalities for youth registration, capturing essential details such as roles, responsibilities, and participation in drilling projects. The system should maintain a comprehensive database of youth members affiliated with the organization.

Teams and Inventory Management:

- ✔ Implement features that allow for the efficient management of drilling teams and inventory. This includes tracking the composition and roles of teams involved in borehole drilling projects, as well as maintaining accurate records of inventory items, such as drilling equipment, pumps, and plumbing materials.

On-Demand Report Generation:

- ✔ Develop a reporting module that enables the generation of comprehensive reports on demand. Users should be able to request reports related to financial summaries, project progress, inventory status, and any other key performance indicators relevant to Uzima Borehole Drilling Company.

UZIMA BOREHOLE SYSTEM

Real-Time Data Processing:

- ✔ Ensure that the system processes data in real-time, providing up-to-date information for decision-making. This includes the immediate recording of youth registrations, team activities, and inventory changes as they occur.

Secure Data Storage:

- ✔ Implement robust data security measures to safeguard sensitive information related to youth members, teams, inventory, and organizational activities. This includes user authentication and authorization to control access to different levels of information within the system.

Scalability and Adaptability:

- ✔ Design the system to be scalable and adaptable to the changing needs of Uzima Borehole Drilling Company. It should accommodate the growth of youth memberships, drilling projects, and inventory items over time.

User Training and Support:

- ✔ Provide training resources and support to users, ensuring they are proficient in utilizing the system's features for data entry, record maintenance, and report generation.

Continuous System Improvement:

- ✔ Establish mechanisms for continuous improvement, gathering feedback from users to enhance system functionalities and address any emerging requirements of Uzima Borehole Drilling Company.

The objectives outlined above aim to guide the development and implementation of the Uzima Borehole Drilling System, ensuring it effectively addresses the specific needs of the organization.

UZIMA BOREHOLE SYSTEM

Advantages.

Time-Saving:

- ✔ The system automates data entry, processing, and reporting, reducing the time required for administrative tasks. This enables Uzima Borehole Drilling Company to operate more efficiently and respond promptly to client needs.

Cost-Efficient:

- ✔ By automating manual processes, the system helps in cost reduction associated with labour, paperwork, and potential errors. This efficiency contributes to overall cost savings for the organization.

Space Efficiency:

- ✔ The digital database eliminates the need for physical storage of extensive paper records, saving floor space and allowing for a more organized and clutter-free working environment.

Accuracy in Processing:

- ✔ Automated data processing ensures a higher level of accuracy in maintaining records, computing fees, and generating reports. This minimizes errors that may occur in manual record-keeping processes.

Corruption-Free:

- ✔ The digital nature of the system reduces the risk of corruption or manipulation of records. Data integrity is maintained, providing reliable information for decision-making and auditing purposes.

Efficiency Improvement:

- ✔ The system streamlines organizational processes, leading to increased overall efficiency. Tasks such as client registration, project coordination, and financial computations are carried out seamlessly, enhancing productivity.

Secure from Damage:

- ✔ Electronic records are less susceptible to physical damage, such as fire or water damage, compared to paper-based records. This ensures the preservation and longevity of important data.

UZIMA BOREHOLE SYSTEM

Secure from Unauthorized Access and Thefts:

- ✔ Robust security measures, including user authentication and access controls, protect sensitive information from unauthorized access or theft. This ensures that only authorized personnel can access and modify specific data within the system.

Neat Work, Timely Processing:

- ✔ The system promotes neat and organized work by providing a structured and standardized platform for data entry and processing. This contributes to timely processing of tasks, preventing delays and enhancing overall operational efficiency.

The adoption of the Uzima Borehole Drilling System brings about multiple advantages, ranging from improved time management and cost efficiency to enhanced data security and accuracy. These benefits collectively contribute to the overall success and effectiveness of Uzima Borehole Drilling Company's operations.

UZIMA BOREHOLE SYSTEM

CHAPTER TWO

FEASIBILITY STUDY

It is the study connected with the first hand before a new system is established in order to establish whether the project is a profit making or loss-making study.

The system analyst carried out a feasibility study and came up with a study that the system will be very effective and will be a profit-making system.

Feasibility Report

The feasibility report indicates that the Uzima Borehole Drilling System is not only effective but also aligns with the organization's goals of financial sustainability. The profitability of borehole drilling operations, coupled with employment creation, makes the system a valuable asset for the organization.

Recommendations:

Based on the positive findings of the feasibility study, it is recommended that the organization proceed with the implementation of the Uzima Borehole Drilling System. This will not only enhance operational efficiency but also position the organization for increased profitability and positive contributions to the community through employment creation.

Cost benefit analysis

The cost of establishing the system and the entries development and operations is as follows:

- ✔ Development cost=Ksh 3,000,000
- ✔ Monthly cost=Ksh 2,000,000
- ✔ Total life time=6 years
- ✔ Total life time cost=n 3000+ (12*200000)6
- ✔ Total cost=Ksh 17,400,000
- ✔ Profit from increased sale per month=Ksh 9,000,000

UZIMA BOREHOLE SYSTEM

@2024PROJECT

BENEFITS

- ✔ Profit from introduction of paperless mails= 2 million
- ✔ Savings from reduced travelling cost=2 million
- ✔ Total monthly benefit= 12million
- ✔ Total life time benefit= 8.64 billion

Schedule feasibility

The system will take 7 Months and every event and timespan was recorded as below

EVENT	PERIOD
1. Problem recognition and definition	4 weeks
2. Information gathering	6 weeks
3. Requirement specification	2 weeks
4. System design	6 weeks
5. System coding	6 weeks
6. System implementation	4 weeks
7. System review and maintenance	Opted
Total	**28 weeks**

UZIMA BOREHOLE SYSTEM

Economic feasibility

Assessment:

An analysis of the economic implications, including costs and potential returns, was conducted. This involved estimating the development costs, ongoing operational expenses, and the expected financial benefits over time.

Outcome:

The economic feasibility study indicated that the system is cost-efficient and has the potential to generate profits. The benefits outweigh the development and maintenance costs, making it a financially viable project.

Operational feasibility

Assessment:

The impact of the proposed system on daily operations, workflow, and staff responsibilities was examined. The system analyst considered how well the new system would integrate into the existing organizational structure.

Outcome:

The operational feasibility study revealed that the proposed system aligns seamlessly with current operations, enhancing efficiency and contributing positively to day-to-day activities.

Technical feasibility

Assessment:

The technical capabilities required for the development and implementation of the database system were evaluated. This included considerations for software and hardware requirements, technical expertise, and compatibility with existing infrastructure.

Outcome:

The study concluded that the technical aspects were feasible, and the organization possesses the necessary resources to support the implementation.

Conclusion

In conclusion, the development of the proposed system is deemed to be a cost-effective initiative that promises to deliver high-quality services. The system is poised to enhance the operational efficiency of the organization, providing a platform for more streamlined processes and improved service delivery. The purpose of the presented report was to apprise members of the challenges faced by the community and to underscore the effectiveness of the new system in addressing these concerns. Despite the anticipated challenges, including the initial cost and maintenance efforts, the system is projected to significantly contribute to the attainment of corporate objectives in the future. The benefits

UZIMA BOREHOLE SYSTEM

it brings, particularly in terms of cost-effectiveness, are expected to outweigh its minimal disadvantages throughout its operational lifetime, making it a worthwhile investment for the organization.

UZIMA BOREHOLE SYSTEM

CHAPTER THREE

FACT FINDING REPORTS

The initiation of the Uzima Borehole Drilling System Management System project has been authorized, and the fact-finding phase is crucial for gathering essential information to inform the development of the new system. The fact-finding activities will involve documentation study, interviews, observation, and questionnaires conducted by members of the group

Members of the group will carry out different fact-finding activities through:

- Documentation study
- Interviews
- Observation
- Questionnaire

Documentation study

Members of the group engaged in a comprehensive documentation study to gather existing information pertinent to the current borehole drilling processes, client registration, project management, financial transactions, and overall organizational operations. This involved reviewing manuals, reports, and any relevant documents to understand the existing system.

Questionnaire

To ensure a comprehensive understanding of various perspectives, questionnaires were distributed to both internal stakeholders (employees) and external stakeholders (clients). The questionnaires were designed to gather specific feedback on the user experience, challenges faced, and suggestions for enhancements in the current processes.

UZIMA BOREHOLE SYSTEM

UZIMA BORE HOLE SYSTEM SPORT CLUB MANAGEMENT SYSTEM

DATE: 20th January 2024

Introduction: The information department is in the process of developing a new system for the above information system. Please help in filling the questionnaire and return to the information technology office before 25th January 2024

Question 1

How do the members feel about the current manual system? Are they satisfied?

☐ Very much ☐ Not at all ☐ A little ☐ No idea

Question 2

How often do you get new members?

☐ Very often ☐ often ☐ not often ☐

Question 3

How long does it take for you to compute the project profits? Elaborate

Your answer

☐ 1 hour ☐ several hours ☐ One day ☐ Many Days

UZIMA BOREHOLE SYSTEM

Interviews

Structured interviews were conducted with key stakeholders, including management, technical staff, administrative personnel, and clients. These interviews aimed to collect insights into their experiences, requirements, and expectations from the new system. Key topics covered included current challenges, desired functionalities, and areas for improvement.

INTERVIEW NAME:	INTERVIEWER NAME:
Date:	
Place:	
Subject:	

TIME ALLOCATED	INTERVIEWER	INTERVIEWEE
3-5 minutes	My name is’”” …’’. I work for a technology department which is in the process of developing a new system and I am here to collect some information. I would like to thank you for availing up here for this interview. Many thanks once again.	
10 minutes	Question one What challenges do you encounter in this system? Explain widely	
10 minutes	Question two	

	Despite the challenges you still trust the organisation and that is why you are here now. What are the benefits of this system?
8 minutes	Question three Once the problems have been identified, what are the possible actions that might be taken to sort out the problems?
6 minutes	Question four How are the profits computed?
6 minutes	Question five. How efficient is the new system in terms of report handling?
1 minute	I would like to thank you and all those who have made this interview a success. The owner of this organization will receive a copy of what has been transpired during this interview.

@2024PROJECT

Observations

Members of the group actively observed the day-to-day operations of Uzima Borehole Drilling Company. This hands-on approach provided valuable insights into the actual workflow, interactions, and challenges faced by staff during borehole drilling projects, client interactions, and administrative processes.

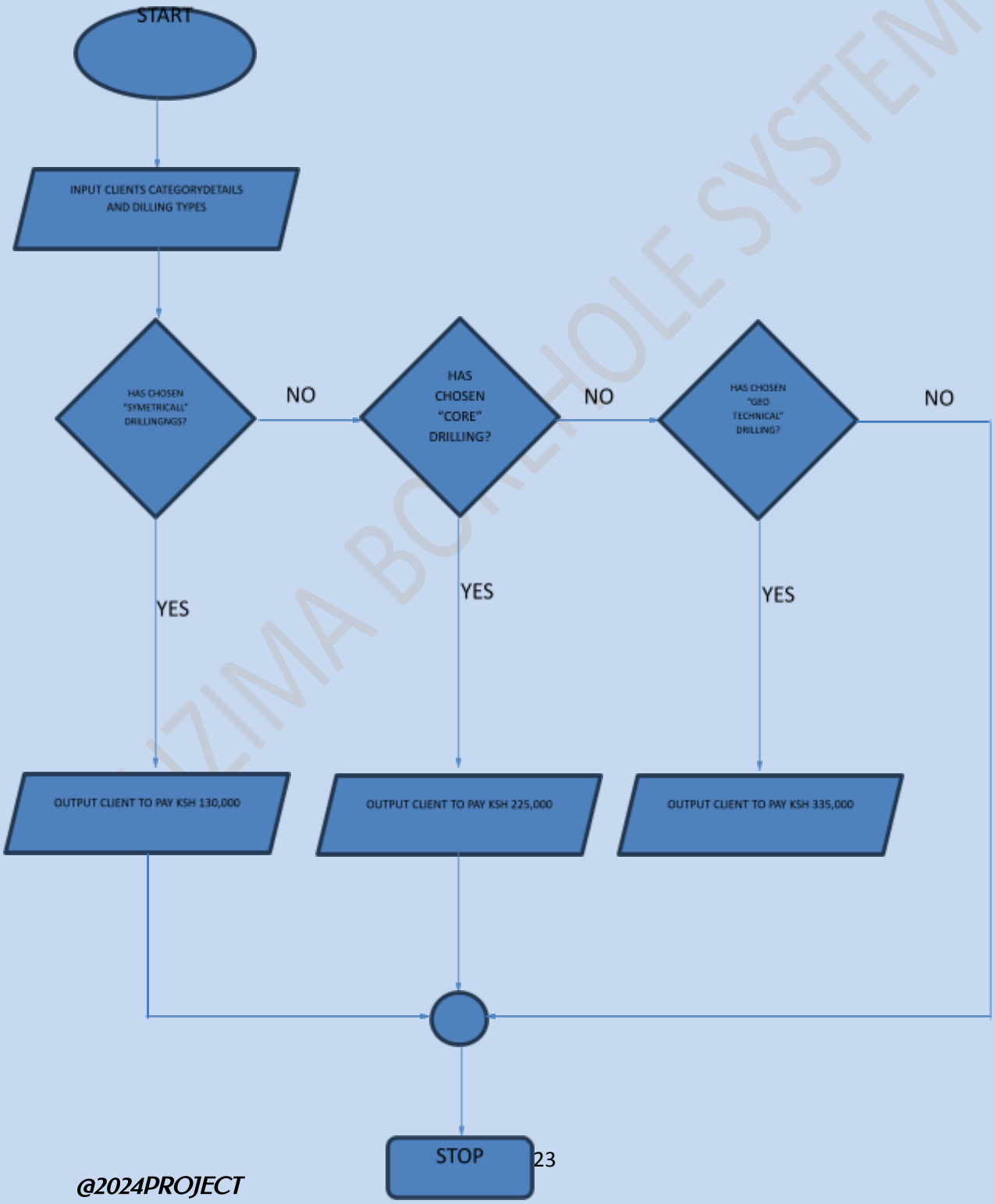
Recommendation

In light of the development and approval of the new Uzima Borehole System Management System by the management, it is strongly recommended for implementation. The new system offers a considerable improvement in user-friendliness compared to the outdated manual system, making it a more efficient and reliable solution. The decision to replace the old system with this computerized alternative is well-founded, considering the enhanced reliability it brings to the organization. While acknowledging that the new system incurs a higher initial cost, its numerous benefits are expected to outweigh the expenses, resulting in more effective and streamlined services for Uzima Borehole Drilling Company. The investment in this modernized system is poised to contribute significantly to the organization's overall efficiency and success in delivering high-quality services.

UZIMA BOREHOLE SYSTEM

CHAPTER FOUR

SYSTEM DESIGN: SYSTEM FLOWCHARTS



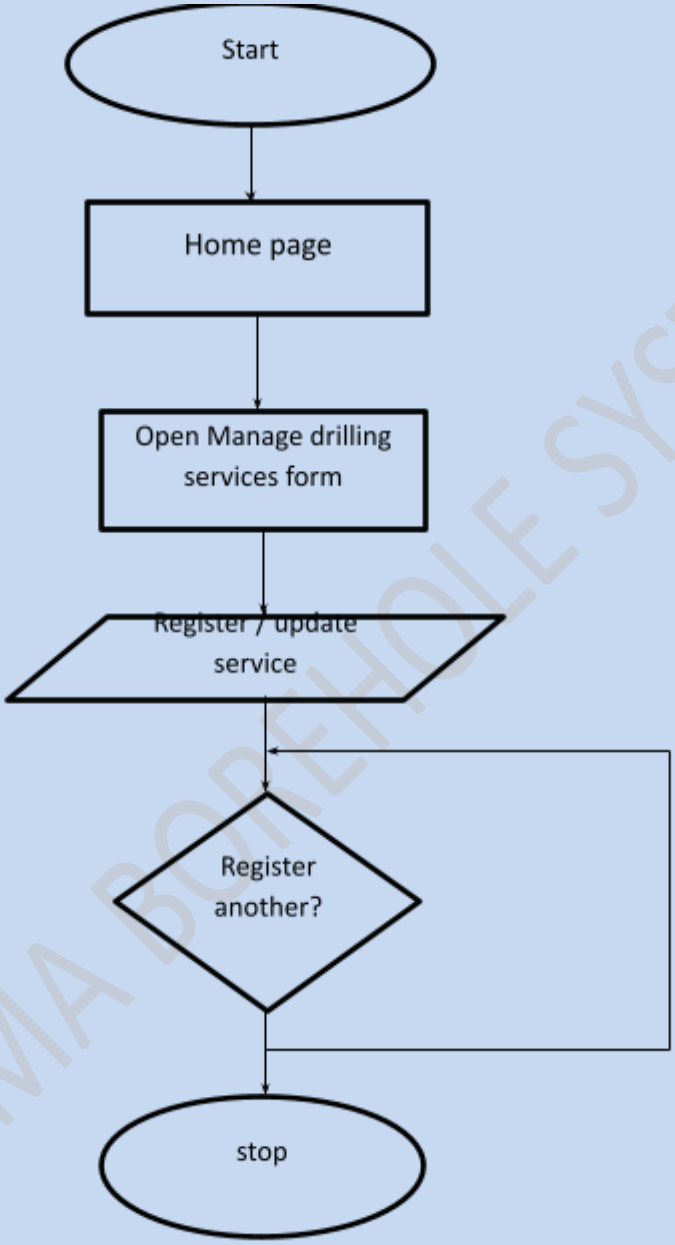
IZIMA BOREHOLE SYSTEM

@2024PROJECT

CHAPTER FOUR

SYSTEM DESIGN: System flowcharts

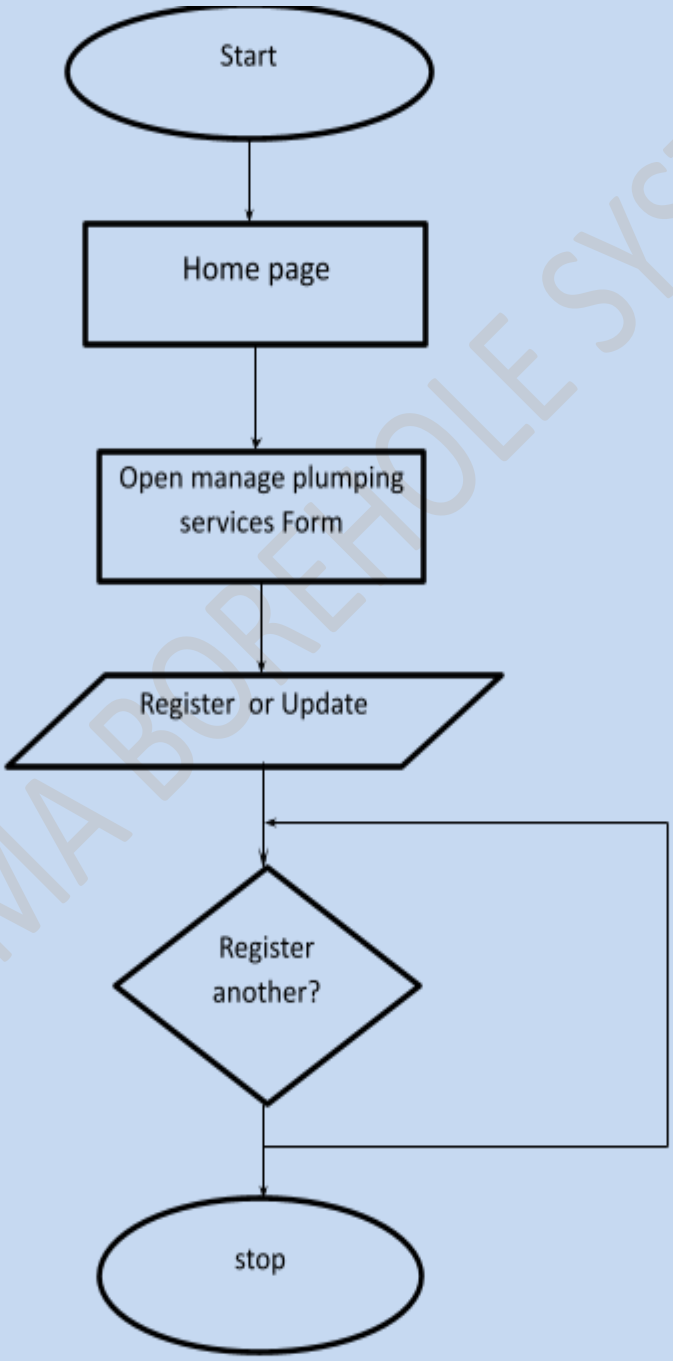
MANAGE DRILLING SERVICES



UZIMA BOREHOLE SYSTEM

@2024PROJECT

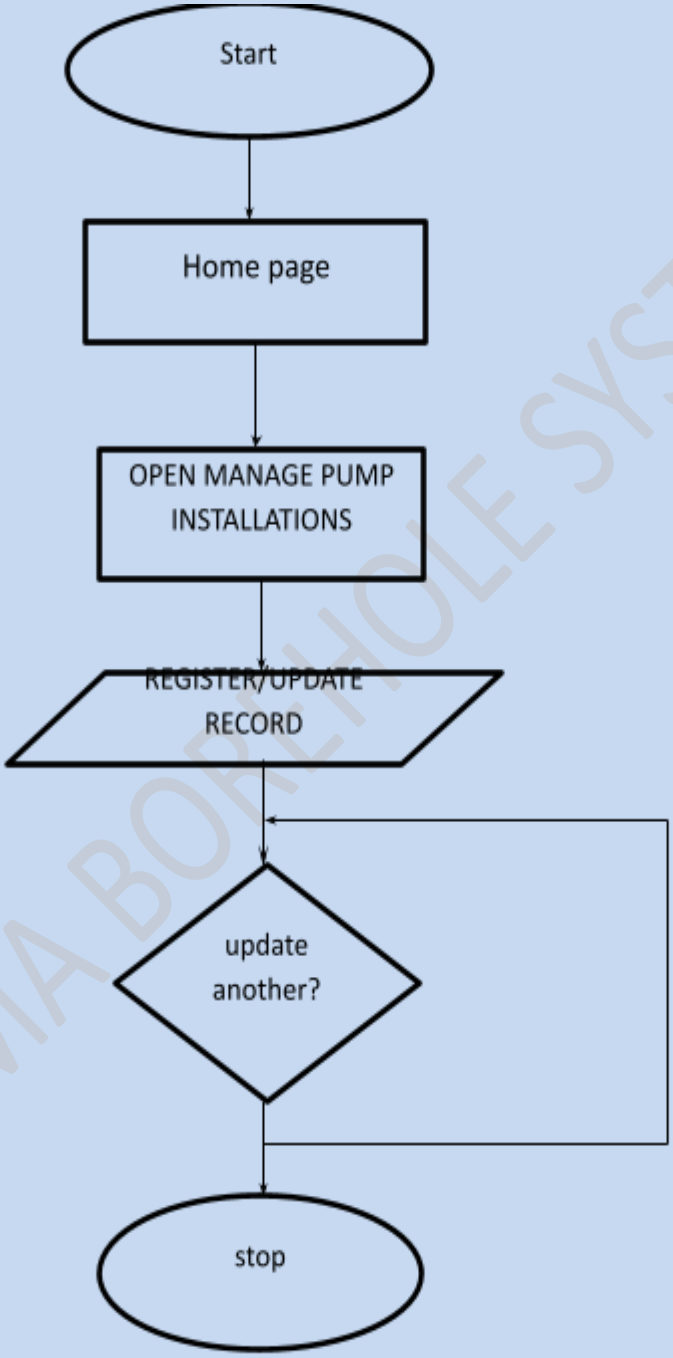
MANAGE PLUMPING SERVICES



UZIMA BOREHOLE SYSTEM

@2024PROJECT

MANAGE PUMP INSTALLATIONS

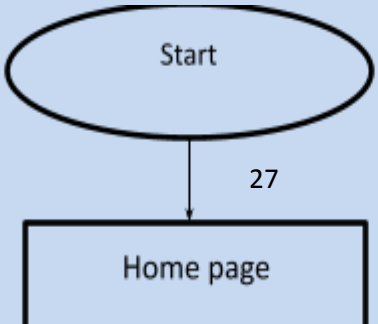


UZIMA BOREHOLE SYSTEM

@2024PROJECT

UZIMA BOREHOLE SYSTEM

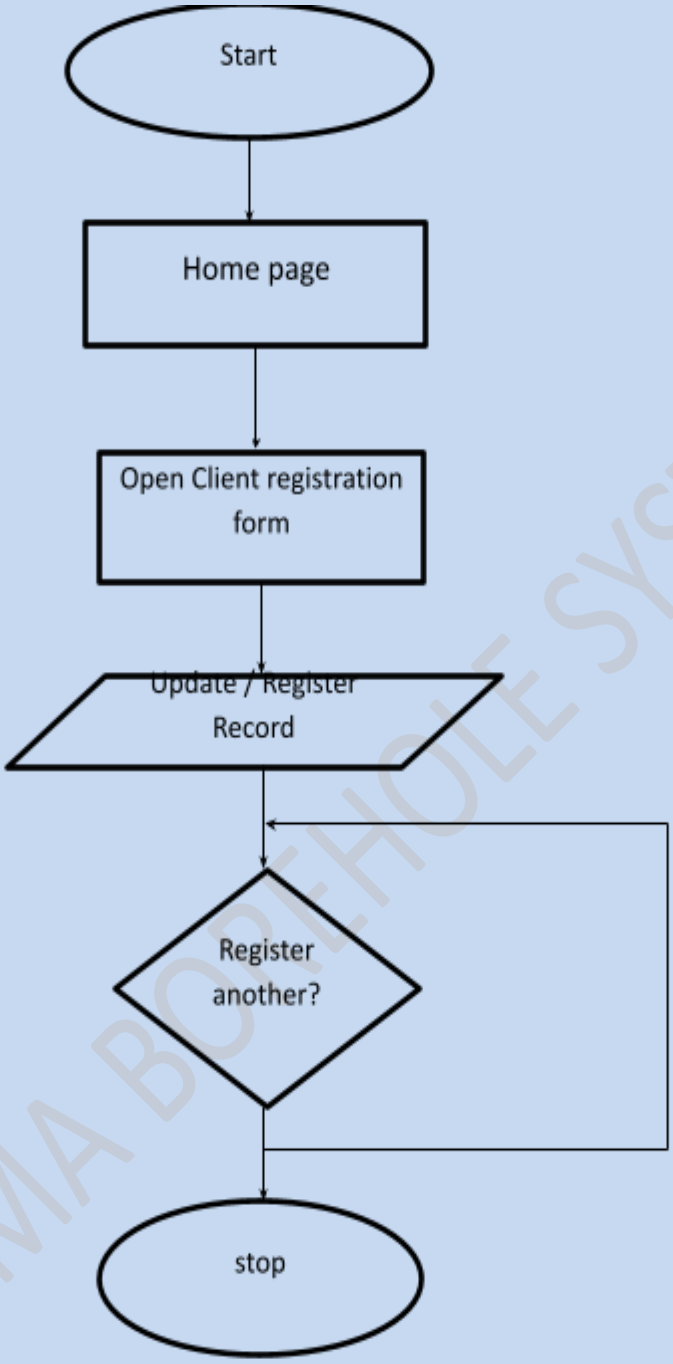
MANAGE TANK PRICES



@2024PROJECT

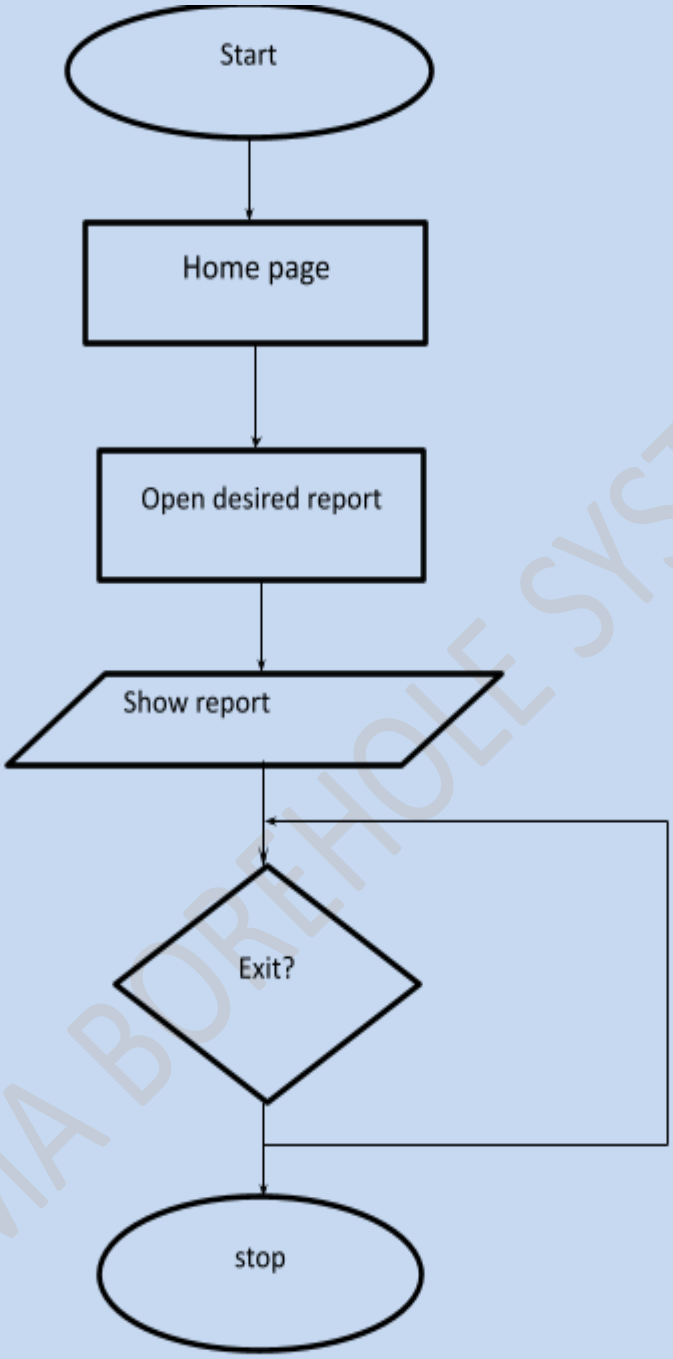
UZIMA BOREHOLE SYSTEM

CLIENT REGISTRATION PORTAL



UZIMA BOREHOLE SYSTEM

OUTPUT FLOWCHAT



UZIMA BOREHOLE SYSTEM

UZIMA BOREHOLE SYSTEM

Figure 1: Clients table

@2024PROJECT

clients

Field Name	Data Type
D	AutoNumber
full_name	Short Text
address	Short Text
telephone_number	Short Text
email	Short Text
borehole_location	Short Text
client_cat	Number
drilling_service	Number
pump_type_link	Number
depth	Number
tank	Number
plumping_plan	Number

UZIMA BOREHOLE SYSTEM

Figure 2. Plumbing installation table

pump_installations

Field Name	Data Type
ID	AutoNumber
pump_type	Short Text
cost	Currency

Figure 3. Plumbing price table

plumping_prices

Field Name	Data Type
ID	AutoNumber
package	Short Text
price	Currency

Figure 4. Services offered table

services_offered

Field Name	Data Type
ID	AutoNumber
service_name	Short Text
down_payment	Currency

Figure 5. Survey local fees table

survey_local_fees

Field Name	Data Type
ID	AutoNumber
client_category	Short Text
survey_field	Currency
local_authority_fee	Currency

Figure 6. Tank prices table

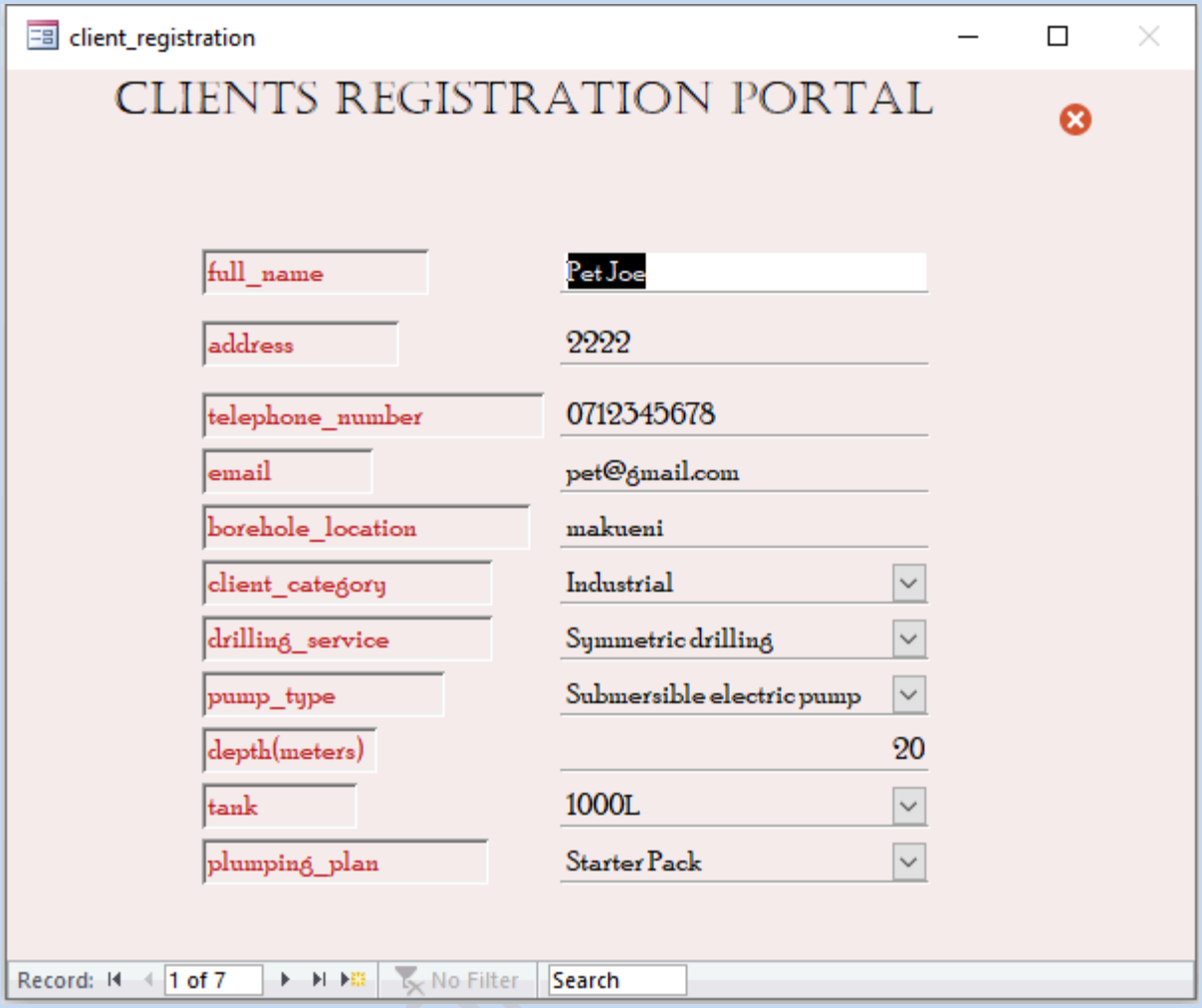
tank_prices

Field Name	Data Type
ID	AutoNumber
capacity	Short Text
cost	Currency

UZIMA BOREHOLE SYSTEM

INPUT DESIGN

FIGURE 7.



UZIMA BO

Figure 8. Drilling services

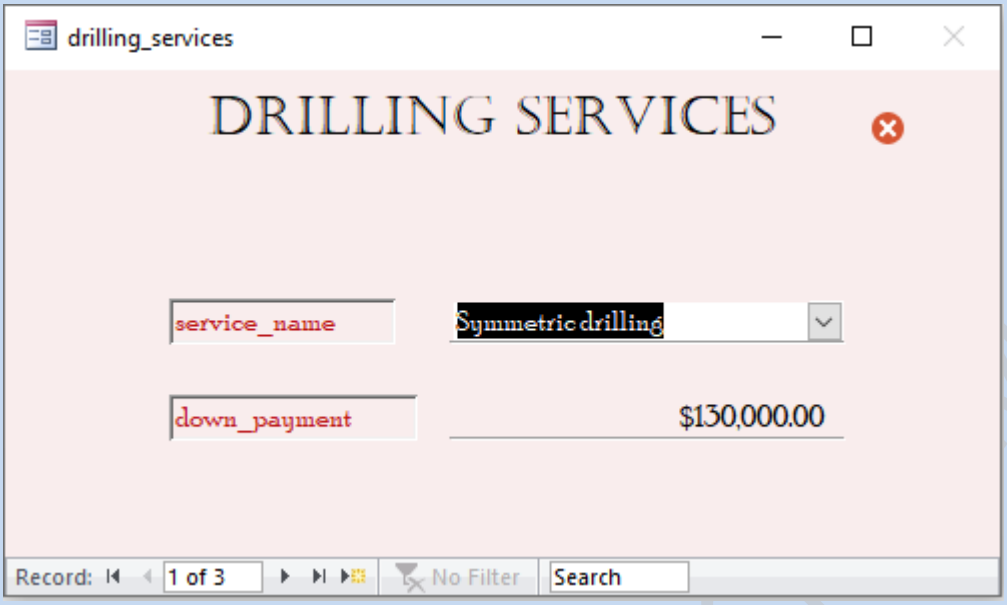
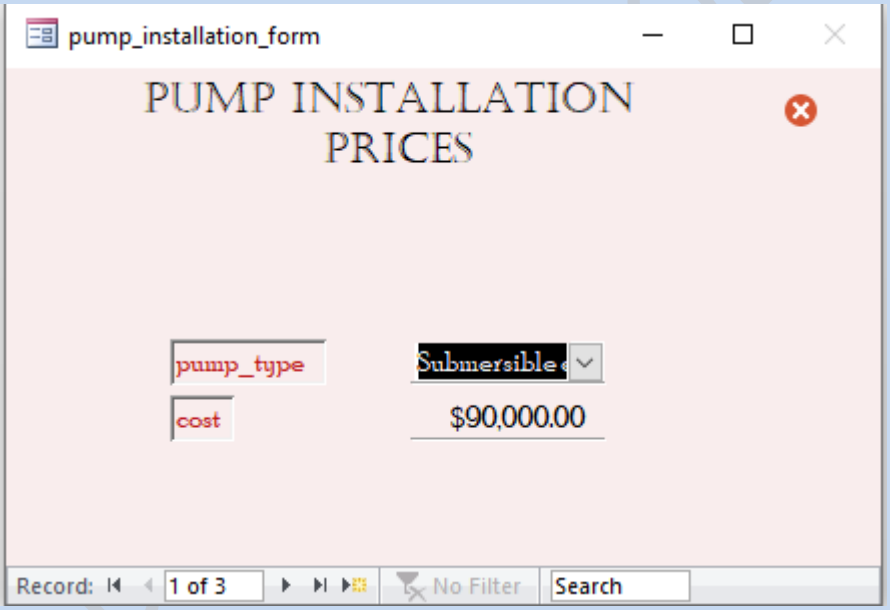


Figure 9. Plumbing installation prices



@2024PROJECT

Figure 10. Plumbing prices

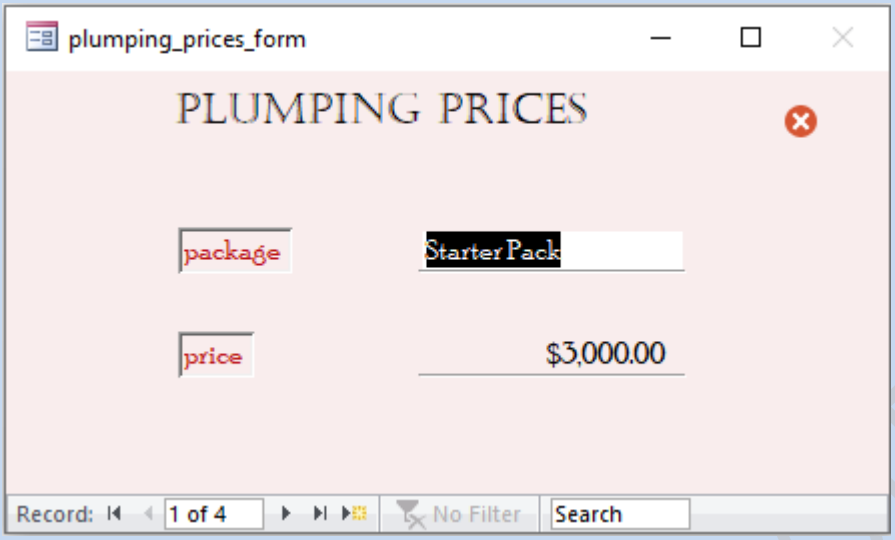
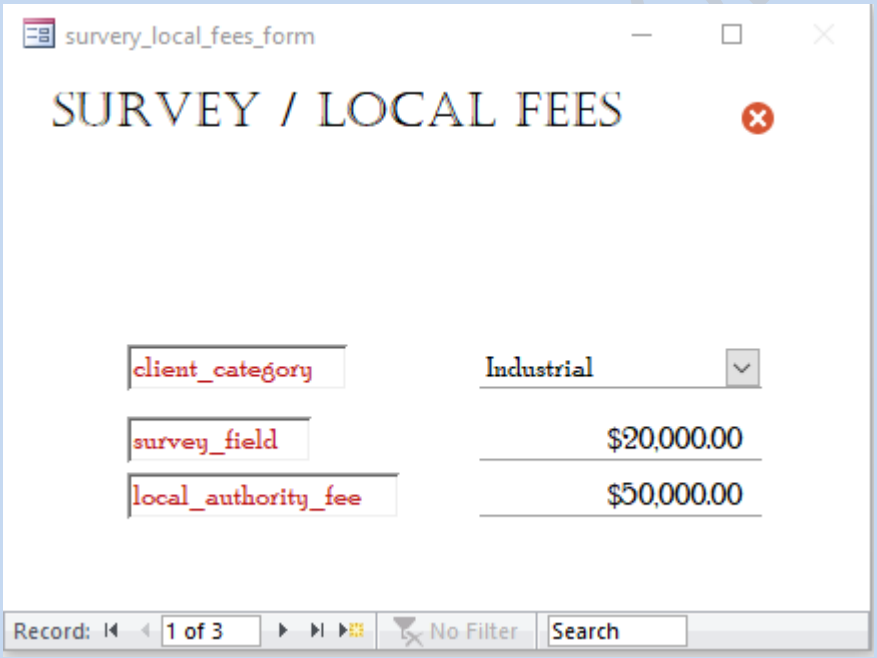
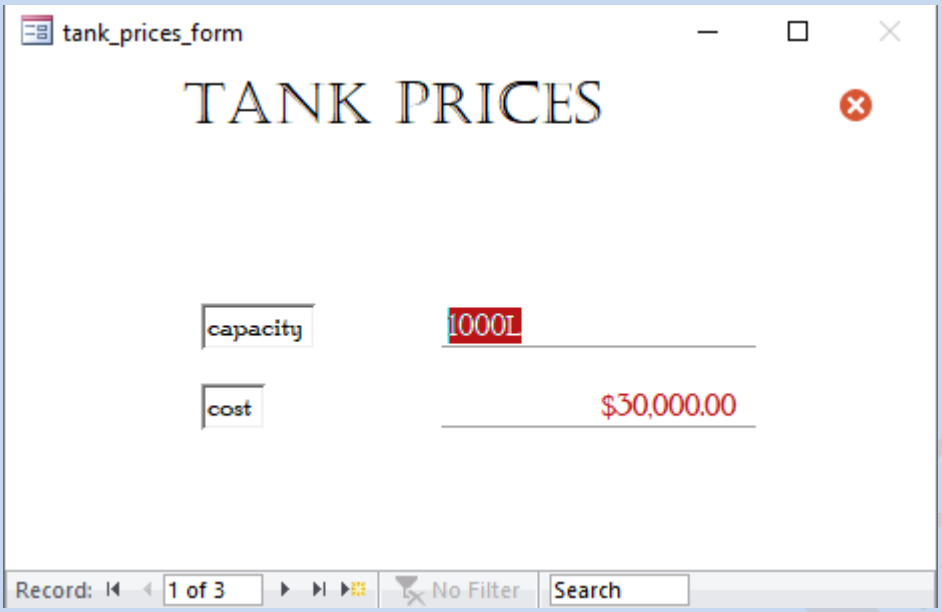


Figure 11. Survey/ local fees



@2024PROJECT

Figure 12. Tank prices form



UZIMA BOREHOLE SYSTEM

4.3 OUTPUT DESIGN

Figure 13. All clients

all_clients

ALL CLIENTS

ID	full_name	address	telephone_number	email	borehole_location	client_category	drilling_service	pump_type_link	depth(m)	tank	plumping_plan
2	Pet Joe	2222	0712345678	pet@gmail.com	makueni	Industrial	Symmetric drilling	Submersible electric pu	20	1000L	Starter Pack
3	Kas Josh	23876	0723541759	kasjosh@gmail.	Kinyambu	Domestic	Core drilling	Submersible electric pu	50	1000L	Premium Pack
4	Jane Doe	5664	0745632847	janedoe@gmail	Naivasha	Commercial	Core drilling	Submersible electric pu	50	2000L	Premium Pack
5	Mark Cuban	4587	0721547862	markcuban@gu	Machakos	Commercial	Geo-Technical drillir	Solar pump	70	2000L	Premium Pack
6	Lebron James	4586	0725698214	lebronjames@g	Kitui	Domestic	Core drilling	Solar pump	1000	1000L	Standard Pack
7	Jianis Antentecumpo	58694	0723654875	jianishimself@g	Wote	Industrial	Geo-Technical drillir	Solar pump	100	2000L	Premium Pack
8	Steve Curry	58964	0721589742	stevecurry@gm	Eldoret	Domestic	Symmetric drilling	Solar pump	25	1000L	Starter Pack

Tuesday, 23 January 2024

Figure 14. Drilling services report.

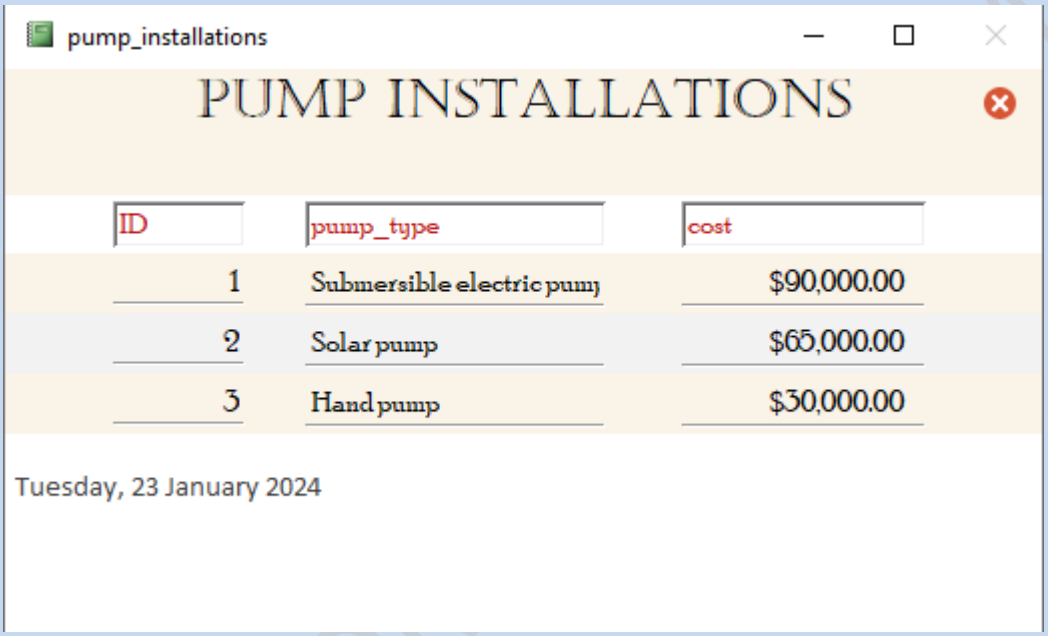
services_offered

DRILLING SERVICES

ID	service_name	down_payment
1	Symmetric drilling	$130,000.00
2	Core drilling	$225,000.00
3	Geo-Technical drilling	$335,000.00

Tuesday, 23 January 2024

Figure 15. pump installation report.



pump_installations

PUMP INSTALLATIONS

ID	pump_type	cost
1	Submersible electric pump	$90,000.00
2	Solar pump	$65,000.00
3	Hand pump	$30,000.00

Tuesday, 23 January 2024

Figure 16. Plumbing packages report

UZIMA BOOKS

@2024PROJECT

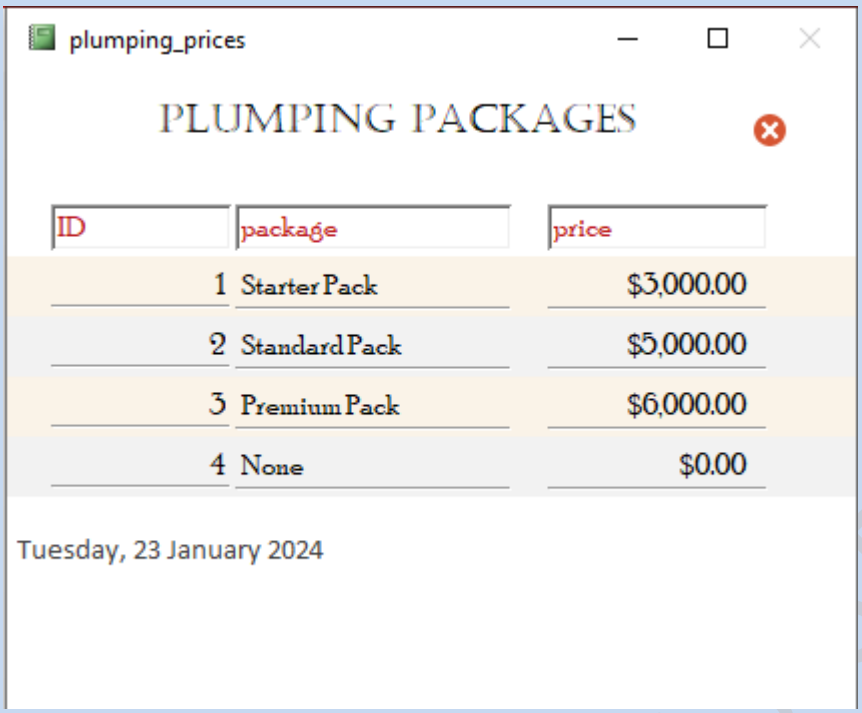
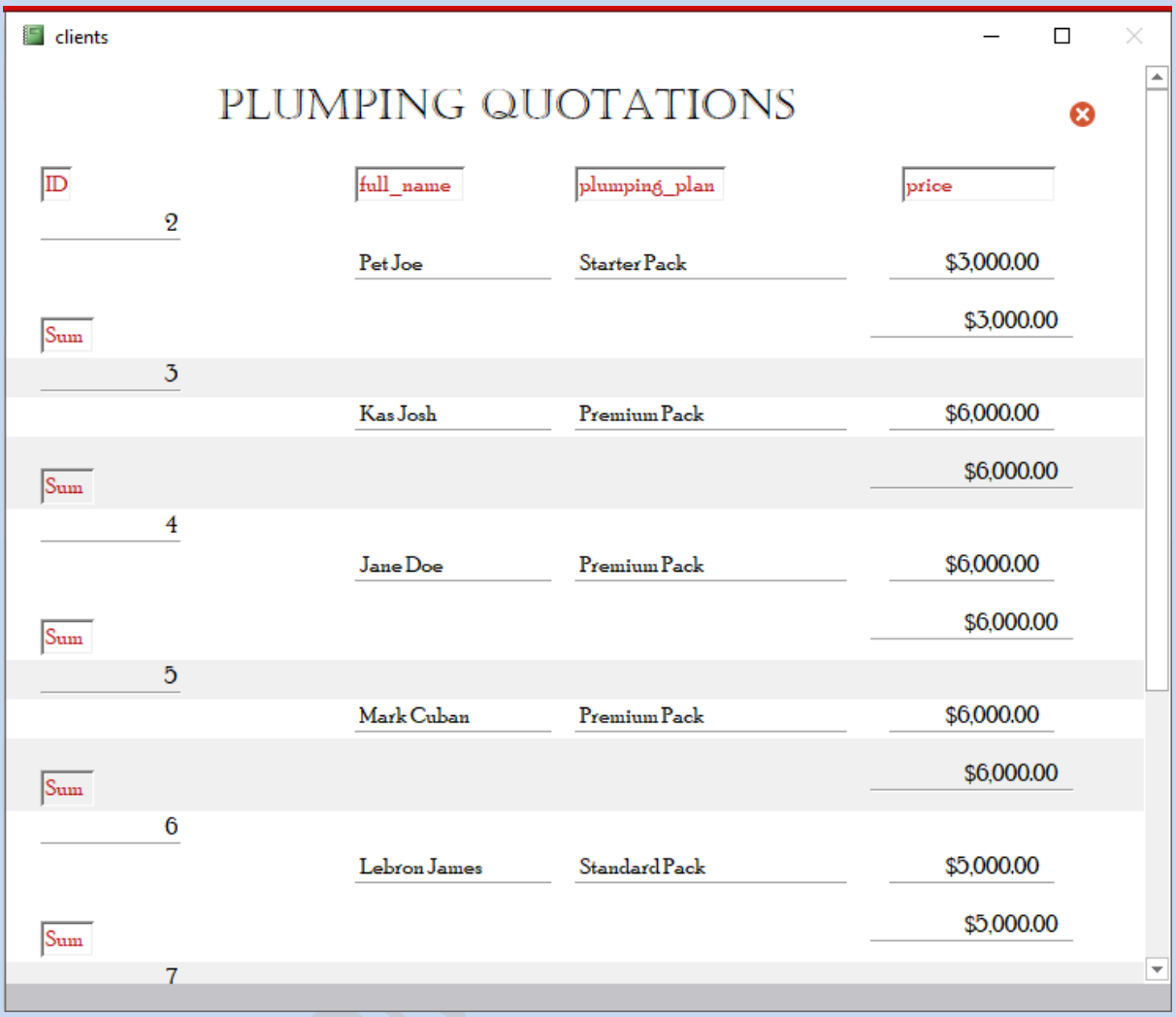


Figure 17. Plumbing quotations.

UZIMA BOREHOLE SYSTEM



clients

PLUMPING QUOTATIONS

ID	full_name	plumping_plan	price
2			
	Pet Joe	Starter Pack	$3,000.00
Sum			$3,000.00
3			
	Kas Josh	Premium Pack	$6,000.00
Sum			$6,000.00
4			
	Jane Doe	Premium Pack	$6,000.00
Sum			$6,000.00
5			
	Mark Cuban	Premium Pack	$6,000.00
Sum			$6,000.00
6			
	Lebron James	Standard Pack	$5,000.00
Sum			$5,000.00
7			

UZIMA BOR

Figure 18. Survey/ loacal authority fees.

survey_local_fees

SURVEY / LOCAL AUTHORITY FEES

ID	client_category	survey_field	local_authority_fee
1	Industrial	$20,000.00	$50,000.00
2	Commercial	$15,000.00	$30,000.00
3	Domestic	$7,000.00	$10,000.00

Tuesday, 23 January 2024

Figure 19. Tank prices report.

tank_prices

TANK PRICES

ID	capacity	cost
1	1000L	$30,000.00
2	2000L	$5,000.00
3	None	$0.00

Tuesday, 23 January 2024

Figure 20. Total revenue generated.

grand_revenue

TOTAL REVENUE

Total Survey Fee	$91,000.00
Total Local Authority	$190,000.00
Total Drilling Amount	$1,605,000.00
Pump Installation Amount	$530,000.00
Cost Per Height	$2,815,000.00
Tanks Revenue	$135,000.00
Plumping Revenue	$35,000.00
Grand Revenue	$4,536,840.00

Tuesday, 23 January 2024 Page 1 of 1

4.5 HARDWARE AND SOFTWARE REQUIREMENTS

Having meticulously reviewed the detailed system design specifications and recommendations, the leadership of the Uzima Borehole Drilling Company has reached a significant decision to embrace a computer-based system. This pivotal choice has set in motion the next phase where the development team is tasked with the responsibility of sourcing all the essential hardware and software resources.

A team of hardware and software specialists within the organization has proactively undertaken visits to multiple suppliers in major towns. Their objective was to obtain comprehensive quotations and pricing details for the necessary resources. Below is a table

detailing the hardware requirements recommendation for the impending implementation of the Uzima Borehole Management System:

No.	Hardware facility	Example	Estimated cost (Ksh)
1	Computer	IBM desktop computer	
2	Storage media	Optical disks, Magnetic disks	Documents collected from various suppliers' quotations
3	POS terminal equipment	Wand reader, Magnetic card reader, Bar code printer	
4	Printers	Laser Printer, Matrix printer	
5	Scanners		
6	others		

This strategic shift towards a computer-based system marks a critical step in enhancing the efficiency and capabilities of Uzima Borehole Drilling Company, aligning with modern technological standards and best practices.

@2024PROJECT

4.7 DATA INTERGRITY CONSTRAINTS

4.8.1 Entity integrity:

Entity integrity, a fundamental concept in database design, establishes essential rules governing the composition of a primary key. These rules apply universally and ensure the integrity and reliability of the database structure. The primary key, being a unique identifier for each record, must adhere to two critical principles.

Firstly, the primary key cannot contain a null value. This enforces a requirement for every record to have a distinct and defined primary key, eliminating ambiguity and contributing to the accuracy of data representation. Secondly, the primary key must be unique across all records in the database. This uniqueness constraint guarantees that no two records share the same primary key value, facilitating precise identification and efficient data management.

In the broader scope of the Uzima Borehole Management System project, these principles of entity integrity become integral in ensuring the robustness and reliability of the database. By adhering to these rules, the system establishes a solid foundation, allowing for precise identification and manipulation of borehole and client records, and contributing to the overall success of the project.

clients

Field Name	Data Type
ID	AutoNumber
full_name	Short Text
address	Short Text
telephone_number	Short Text
email	Short Text
borehole_location	Short Text
client_cat	Number
drilling_service	Number
pump_type_link	Number
depth	Number
tank	Number
plumping_plan	Number

Referential integrity:

Referential integrity is a fundamental principle in database design that governs the relationships between tables, particularly in a one-to-many relationship between entity tables. This principle ensures the consistency and accuracy of data across related tables in a relational database.

In the context of a one-to-many relationship, referential integrity entails the following key aspects:

Foreign Key Constraint:

- In a one-to-many relationship, one table (the "many" side) typically contains a foreign key that references the primary key of another table (the "one" side). This foreign key establishes a link between the two tables.

Uniqueness of the Primary Key:

- The primary key in the "one" side must be unique to ensure each record can be uniquely identified. This uniqueness is maintained by the primary key constraint.

Consistency in Foreign Key Values:

- The foreign key values in the "many" side must correspond to existing values in the primary key column of the "one" side. This ensures that relationships between records are valid and that every reference in the "many" side points to a valid record in the "one" side.

Cascade Actions:

- Referential integrity often involves defining cascade actions, such as "CASCADE DELETE" or "CASCADE UPDATE." These actions dictate what happens to related records in the "many" side when a corresponding record in the "one" side is deleted or updated.

By enforcing referential integrity, a database system ensures that relationships between records are valid, and it prevents situations where there are "orphaned" records in the

UZIMA BOREHOLE SYSTEM

"many" side with no corresponding record in the "one" side. This contributes to the overall consistency and reliability of the database.

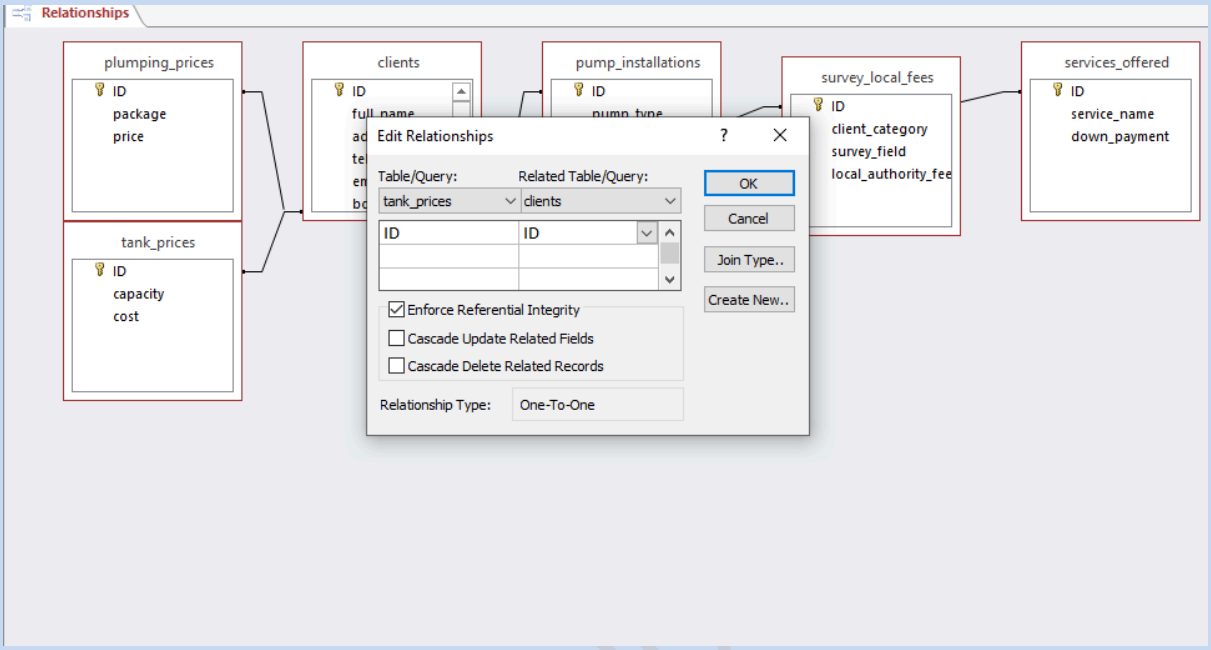


Figure 23 Relationships

UZIMA BOREHO

CHAPTER FIVE

SYSTEM IMPLEMENTATION

Staff training

In the process of system implementation, staff training emerges as a pivotal component to ensure a seamless transition and optimal utilization of the Uzima Borehole Management System. Recognizing the significance of well-prepared staff in the successful adoption of the system, our approach emphasized accessibility to comprehensive documentation, particularly user manuals, to facilitate efficient training.

The availability of user manuals significantly contributes to the ease, speed, and effectiveness of staff training. The risk of system implementation failure is mitigated when staff members are adequately trained, thereby preventing potential losses of company resources. To achieve this, the following methods were employed:

Demonstrations:

An integral aspect of our training strategy involved conducting demonstrations. This approach aimed to provide clear and easily understandable insights into the functionalities of the system. Through hands-on demonstrations, staff members were guided on how to navigate and utilize the system effectively.

Workshops:

Complementing the demonstration sessions, we organized workshops to further enhance the practical application of the knowledge imparted during training. By exposing trainees to real-world scenarios and challenges, we aimed to reinforce their understanding and competence in using the Uzima Borehole Management System.

Manuals:

In addition to interactive sessions, we prioritized the provision of comprehensive manuals to each staff member. These manuals served as valuable reference materials, allowing users to revisit and reinforce their understanding of the system at their own pace. The inclusion of manuals aimed to support continuous learning and serve as a quick reference guide during daily operations.

UZIMA BOREHOLE SYSTEM

@2024PROJECT

By integrating these training methods, we strived to empower the staff with the necessary skills and knowledge required for proficient use of the Uzima Borehole Management System. This multifaceted training approach was designed not only to impart theoretical understanding but also to instil practical competence, ensuring a smooth and successful implementation process.

Changeover strategy

The changeover strategy plays a pivotal role in facilitating the transition from the old system to the new system, determining how the organization effectively adopts and starts utilizing the upgraded technology. In the Uzima Borehole Management System implementation, we employed a Parallel Changeover strategy to ensure a smooth and controlled transition.

Parallel changeover involves running both the old and new systems concurrently for a certain period. This overlapping phase allows users to operate in both environments until they gain confidence in the new system, subsequently leading to the phased-out discontinuation of the old system. This gradual shift minimizes disruptions and offers several advantages and disadvantages.

Advantages:

One notable advantage of the parallel changeover strategy is its ability to lower the risk to business operations. By allowing the simultaneous operation of both systems, thorough testing and validation of the new system occur in a real-world environment. Users can gradually acclimate to the new features, functionalities, and workflows, reducing the likelihood of critical issues disrupting daily operations.

Disadvantages:

However, this method comes with certain drawbacks, primarily in terms of cost. Running both the old and new systems in parallel necessitates the engagement of additional resources to ensure the smooth operation of the new system. This includes additional hardware, software, and human resources, contributing to increased expenses during the transitional period.

UZIMA BOREHOLE SYSTEM

Despite the associated costs, the parallel changeover strategy is chosen for its ability to provide a safety net during the transition, allowing for thorough testing and user adaptation. The decision to implement this strategy in the Uzima Borehole Management System reflects our commitment to minimizing risks and ensuring a successful integration of the new system into the existing operational landscape.

File conversion

File Conversion in System Implementation:

In the course of implementing a new system, the modification or change in the format of data files becomes a crucial consideration. The Uzima Borehole Management System is no exception, and as it operates presently, there is an indication that a transition to a new operating system could significantly enhance its efficiency. Unlike the need for a complete overhaul of hardware systems, the improvement is anticipated through the installation of new application software, thereby presenting a software-centric solution.

Operating System Enhancement:

The current operational dynamics suggest that the Uzima Borehole Management System would benefit from a transition to a more advanced operating system. This change aims to optimize system efficiency, responsiveness, and overall performance. It is essential to note that this enhancement primarily revolves around a software upgrade rather than necessitating extensive changes to the existing hardware infrastructure.

Software Replacement Approach:

To achieve the desired efficiency, the proposed solution involves the installation of new application software and the customization of existing database application software. This strategy eliminates the need for hardware replacements, making the transition cost-effective and minimally disruptive to the ongoing operations. The focus lies in leveraging advanced software capabilities to enhance the functionality and responsiveness of the Uzima Borehole Management System.

UZIMA BOREHOLE SYSTEM

This file conversion and software-centric approach ensure a seamless integration of the new system, allowing for enhanced data processing, improved user experience, and overall system optimization without the need for extensive hardware modifications. The emphasis remains on leveraging innovative software solutions to propel the system's efficiency to new heights.

UZIMA BOREHOLE SYSTEM

CHAPTER SIX

6.1 USER MANUAL

6.1.1 System installation

HOW TO INSTALL THE SYSTEM ON YOUR COMPUTER:
Follow these simple steps to install the Uzima Borehole Management System on your computer:

- ✔ **Load the Program:**

Begin by clicking on the Start Menu located on your computer's taskbar.

- ✔ **Access Microsoft Word:**

Once the Start Menu is open, navigate and click on Microsoft Word to access the application.

- ✔ **Choose Uzima Borehole System:**

Within Microsoft Word, locate and choose the Uzima Borehole System Sport Club System from the list of available options.

- ✔ **Completion:**

After selecting the Uzima Borehole System, follow any on-screen prompts or instructions to complete the installation process.

- ✔ **Assurance:**

Once the installation is complete, you can be assured that the Uzima Borehole Management System is now successfully installed on your computer.

By following these straightforward steps, you can quickly and efficiently install the Uzima Borehole Management System, enabling you to access its features and functionalities seamlessly. If you encounter any issues during the installation process, refer to the system documentation or contact our support team for assistance.

UZIMA BOREHOLE SYSTEM

6.1.2 Running the interface

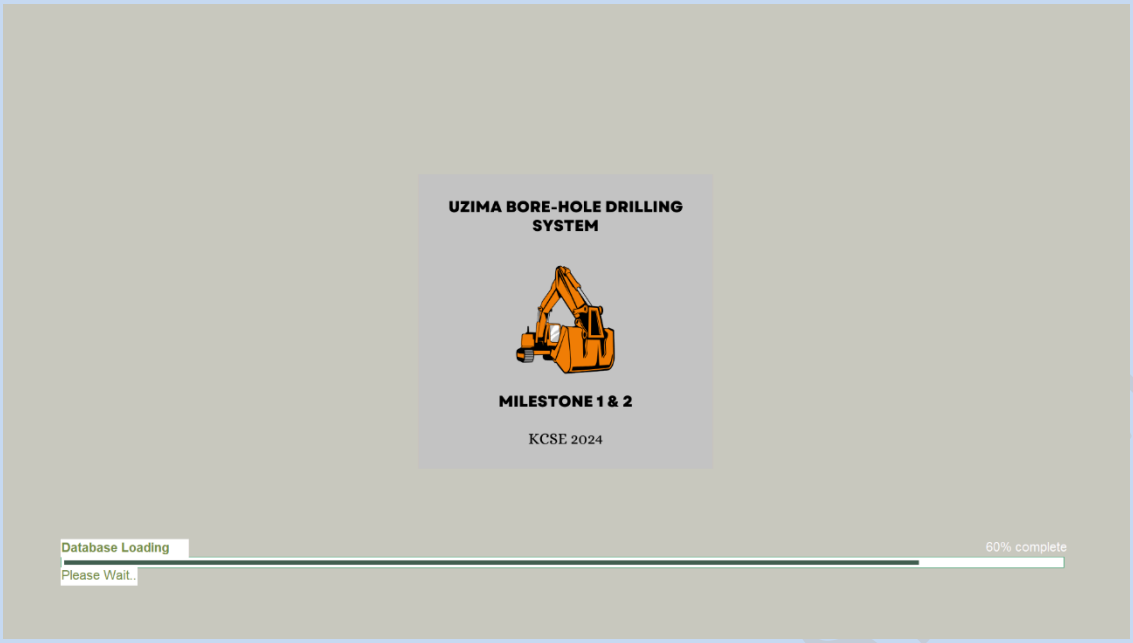


FIGURE 26

This is the home page from which all the user activities will be carried out.

From here you can open forms, close forms close database and open reports.

To close open forms, click the buttons or the icons.

To close the forms and the reports, click the home button.

To go back to the previous screen , click the back button icon.

UZIMA BOREHOLE S

6.1.4 Error handling

member registration | member registration : Image9 : On Click

```
OnError
        Go to   Next
  Macro Name
GoToControl
 Control Name   =[Screen].[PreviousControl].[Name]
ClearMacroError

⊟ If   Not [Form].[NewRecord]   Then
    RunMenuCommand
        Command   DeleteRecord
End If

⊟ If   [Form].[NewRecord] And Not [Form].[Dirty]   Then
    Beep
End If

⊟ If   [Form].[NewRecord] And [Form].[Dirty]   Then
    RunMenuCommand
        Command   UndoRecord
End If

⊟ If   [MacroError]<>0   Then
    MessageBox
```

Macros to handle errors when they occur in the system have been developed. For instance when an error occurs when the user clicks the delete button, proper macros have been put in place to handle that exception.

CHAPTER SEVEN

7.0 BIBLIOGRAPHY

The following books and reference materials were instrumental in the development of the Uzima Borehole Management System:

1. Taylor G: "GCSE Computer Studies," Macmillan Press Limited, 1991.
2. Microsoft Access 2000: "Northwind Sample Database," 1999.
3. Saleemi N.A: "Systems Theory and Management Information Systems," N.A. Saleemi Publishers, 1997 & 2000.
4. Jeffrey L.W. ET al: "System Analysis and Design Methods," McGraw-Hill, 2000
5. Sarah E.H, Glen J. C: "Microsoft Access 2000 Complete Edition," McGraw-Hill, 2000.
6. Williams BK: "Using Information Technology: A Practical Introduction to Computers and Communications," IRWIN.

These references have played a crucial role in shaping the conceptual framework, design, and implementation of the Uzima Borehole Management System, providing valuable insights and expertise in computer studies, systems theory, management information systems, system analysis and design methods, and practical applications of information technology.

UZIMA BOREHOLE SYSTEM

@2024PROJECT

CONCLUSION

All these discoveries affirm that the Uzima Borehole System stands out as the optimal choice for implementation in an organization related to borehole drilling services. This preference arises from its straightforward installation process, cost-effectiveness at the outset, and, most importantly, its exceptional proficiency in report generation and table creation.

The Uzima Borehole System excels in its simplicity of installation, ensuring accessibility for a broad user base. The straightforwardness of the installation contributes to the user-friendliness of the system, facilitating a swift and uncomplicated adoption process.

Affordability is a significant advantage of the Uzima Borehole System. The system offers a cost-effective solution for organizations, enabling them to initiate operations without incurring prohibitive expenses.

Critical to its acclaim is the system's efficiency in producing reports and crafting tables. It effortlessly calculates member contributions, tracks the highest purchase amounts made by clients, and accurately computes the services rendered. The system's complete computerization enhances its ease of use, making it accessible to a diverse range of users.

In summary, the Uzima Borehole System emerges as the epitome of an efficient, cost-effective, and user-friendly solution tailored to meet the unique needs of organizations involved in borehole drilling services. Its ability to seamlessly calculate financial contributions, monitor purchases, and manage services positions it as the top-tier system for organizations seeking a reliable and comprehensive management solution.

UZIMA BOREHOLE SYSTEM

END OF MY PROJECT